STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Negligible molecular volume and forces except during elastic collisions.

Q2 - 1 mark

pV=nRT.

Q3 - 1 mark

3k_BT/2.

Q4 - 1 mark

Molecular momentum transfer to walls.

Q5 - 1 mark

Three translational.

Q6 - 1 mark

It increases.

Q7 - 1 mark

3R/2.

Q8 - 1 mark

sqrt(3RT/M).

Q9 - 2 marks

Doubles.

Q10 - 2 marks

About 300 K.

Q11 - 2 marks

approx 6.21x10^-21 J.

Q12 - 2 marks

2.

Q13 - 2 marks

3nRT/2.

Q14 - 2 marks

Negligible particle volume and negligible intermolecular forces except collisions.

Q15 - 3 marks

approx 517 m/s.

Q16 - 3 marks

approx 249 J.

Q17 - 3 marks

One-third.

Q18 - 3 marks

Three translational modes contribute 3k_BT/2 per molecule.

Q19 - 4 marks

Lighter gas speed is twice as high.

Q20 - 4 marks

U=(5/2)(2)(8.31)(300)=24,930 J approx 24.9 kJ.